DUNGEON OF ETERNITY

Group-22
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1. Introduction to the Gameplay

GameStudio's immersive role-playing game Dungeon of Eternity immerses players in the fantasy universe. In this game we have this magical world called “Endless” and there is a dungeon known as “Noend”. This is a mysterious dungeon with magical power and treasure and it’s also filled with dangerous monsters and traps. No one knows who created this dungeon, no one knows how far they can go. The deeper you go, the harder and stronger you become.
The game allows the player to choose and develop characters. Each player starts from level one, as they adventure the dungeon, they can level up, earn XP, and upgrade their equipment. The game has different characters of adventures that every player can play with like a hero with swords, a lancer with a spare, an archer with an arrow, a warrior with a hammer, a medic with healing magic, a mage with magical power, a tank with the shield.

In this game, we have this adventurer's guild, The guildhall, where every signed-up player can gather and make a party with other players. They can go to the dungeon adventure together only if they have a subscription to the game. Every Signed-up Player can come to the guild to play dots and cards for free. They can earn game currency and pay for the subscriptions.

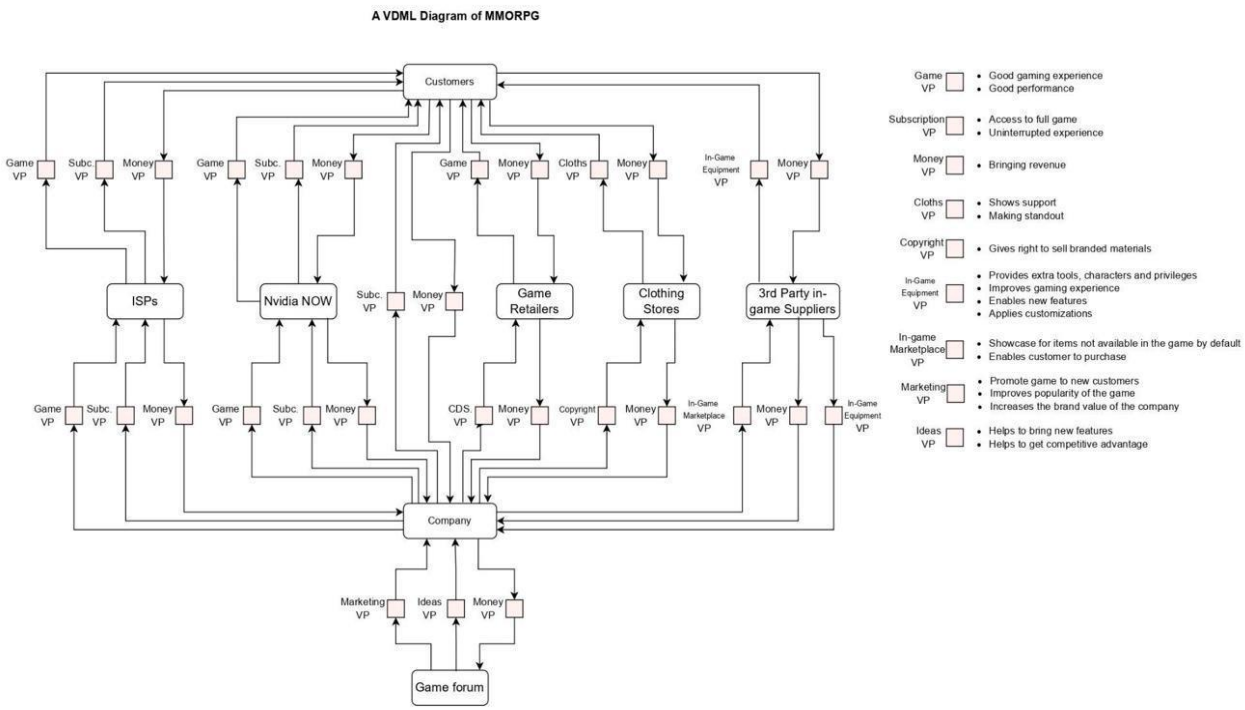
This guildhall has a marketplace, where players can buy and sell their equipment to other players. Also near the guild Hall, there is an armor shop, weapons smith, tailor shop, and medical shop. Here every Player can upgrade their equipment.
The dungeon is divided into levels and each one presents different challenges. The deeper you go the more dangerous the enemy becomes, monsters become stronger, traps become deadlier, and boss-like magical creatures appear. Clearing the levels earns you XP, gems, and rear materials, which players can use for trading.

Introduction of the business plan

Our company’s main target is to make profits by selling subscriptions. Everyone who purchases the CD version of the game receives a key patch that includes the game and offers either 6 or 12 months of subscription, depending on the price they are paying. And those accessing the game through Nvidia NOW or their ISP get a 7-day free trial of the subscription. Everyone can join the game, enter the guildhall and socialize with other players, can play dots and cards but only the subscription holder can explore the dungeon “NoEnd”. Every signed-up player of the game can earn in-game currency by playing dots and cards inside the guildhall of the game and later with this currency, they can get a subscription to the game as well. Third-party brands can also get access to this guildhall and promote their brands. But they must have an agreement with our company.

2. Value Network Analysis

2.1 VDML Diagram



2.2 VDML Diagram Explanation

The VDML (Value Delivery Modeling Language) diagram of the Dungeon of Eternity game illustrates various value exchanges between the agents involved in the game ecosystem.

1. Customers

- **Role**: Players of the game.
- **Value Provided to the company**:
  - **Money VP**: Players contribute money through subscriptions and in-game purchases.

- **Value Received from the company**:
  - **Game VP**: Players receive a possibility to play the game.
  - **In-Game Equipment VP**: Players can purchase or acquire in-game equipment to enhance their characters.
  - **Subscription VP**: Subscribers get uninterrupted access to the full game, which offers a better experience.

2. Nvidia GeForce NOW, ISPs and local CD stores

- **Role**: Distributors of digital and physical copies of the game.
- **Value Provided to the company**:
    - **Money VP**: pay money for the rights to distribute the game and subscriptions to the company.

- **Value Received from the company**:
    - **Subscription VP**: They can generate revenue by selling game subscriptions to their customers
    - **Game VP**: They can generate revenue by selling the game to their customers

3. Game Forum

- **Role**: A platform where players can gather, share ideas, and provide feedback.
- **Value Provided to the company**:
    - **Ideas VP**: The forum allows for the exchange of player ideas, feedback, and suggestions that contribute to the game's continuous development.
    - **Marketing VP**: The forum serves as a promotional tool that enhances the game's visibility and popularity.

- **Value Received from the company**:
    - **Money VP**: Receive money from the company as a sponsorship.

4. 3rd-Party In-Game Suppliers

- **Role**: External entities providing additional in-game items.
- **Value Provided to the company**:
    - **In-Game Equipment VP**: These suppliers offer exclusive in-game items that players can buy to enhance their gameplay.
    - **Money VP**: Pay money for in-game marketplace access to the company.
- **Value Received from the company**:
    - **In-Game Marketplace**: receive in-game marketplace access from company which can be used to generate revenue

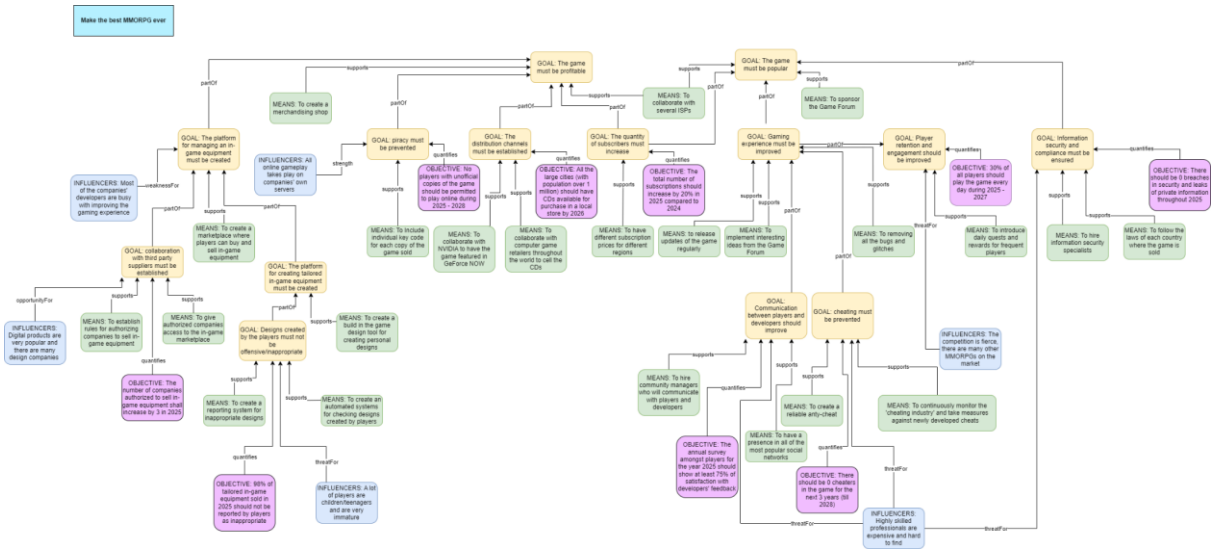
5. Merchandise (Clothing Stores)

- **Role**: External stores that sell branded merchandise tied to the game.
- **Value Provided to the company**:
    - **Money VP**: Pay for copyright to the company.

- **Value Received**:
    - **Copyright VP**: They gain the rights to sell branded in-game materials which are licensed from Game Studio.

3. Goal Design

3.1 Goal Model



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3.2 Goal Model Explanation

The MMORPG project's Goal Model provides an organized summary of the main goals, objectives, and means that are essential to the game's success

3.2.1. Top-Level Goals:

i. **The Game Must Be Profitable**:
Ensuring profitability is a primary concern. It is supported by several subgoals and means.

ii. **The Game Must Be Popular**:
Popularity drives player engagement and attracts new players. The game has a free-to-play element (people can play parts of the game without any subscription), so making the game popular is essential for future success.

3.2.2. Sub-Goals

These are essential building blocks that contribute to achieving the main objectives:

i. **Player Retention and Engagement Should Be Improved**:
The game's capacity to maintain player engagement over time will determine how long it lasts. Daily quests, frequent updates, and improved player-developer communication all help to encourage this.

ii. **The Quantity of Subscribers Must Increase**:
Profitability depends on expanding the player base. Two ways to accomplish this are by providing varying subscription costs for various geographical areas and by enhancing the gaming experience.

iii. **Gaming Experience Must Be Improved**:
Maintaining current gamers and drawing in new ones both depend on a smooth, enjoyable gaming experience. Regular updates, bug fixes, and working with ISPs to improve game speed are how this is accomplished.

iv. **Communication Between Players and Developers Should Improve**:
Better communication makes gamers feel heard and creates a feeling of community, both of which can increase engagement and loyalty. Two important tactics include sponsoring gaming forums and employing community managers.

v. **Information Security and Compliance Must Be Ensured**:
|Ensuring legal compliance and maintaining player data are essential. Important parts of this procedure include hiring security experts and following the local laws in the areas where the game is sold.

vi. **Cheating Must Be Prevented**:
A fair and pleasurable gaming experience depends on maintaining the game's integrity by prohibiting cheating. Developing an anti-cheat system and keeping an eye on emerging cheating techniques are part of the model.

vii. **The Platform for Managing In-Game Equipment Must Be Created**:
To organize and personalize their in-game gear, players require an in-game marketplace where players can purchase and sell goods.

viii. **Collaboration with Third-Party Suppliers Must Be Established**:
The value of the game is increased by working with outside vendors to provide unique and customized in-game items.

ix. **The distribution channels must be established**:
Expanding the game's reach across various platforms and geographical areas is essential for boosting its appeal and financial success.

x. **The platform for creating tailored in-game equipment must be created**:
This boosts player engagement by enabling them to personalize their experience. Designs created by players must not be inappropriate or offensive to other players.

3.2.3. Means, objectives, influencers

- **Means**

The goal model incorporates lots of means that show which actions must be performed in order to achieve previously defined goals. Every leaf goal has at least one mean.

- **Objectives**

Objectives are also present in order to quantify these goals. Each objective has a defined time period in which it should be achieved. Every leaf goal has an objective.

- **Influencers**

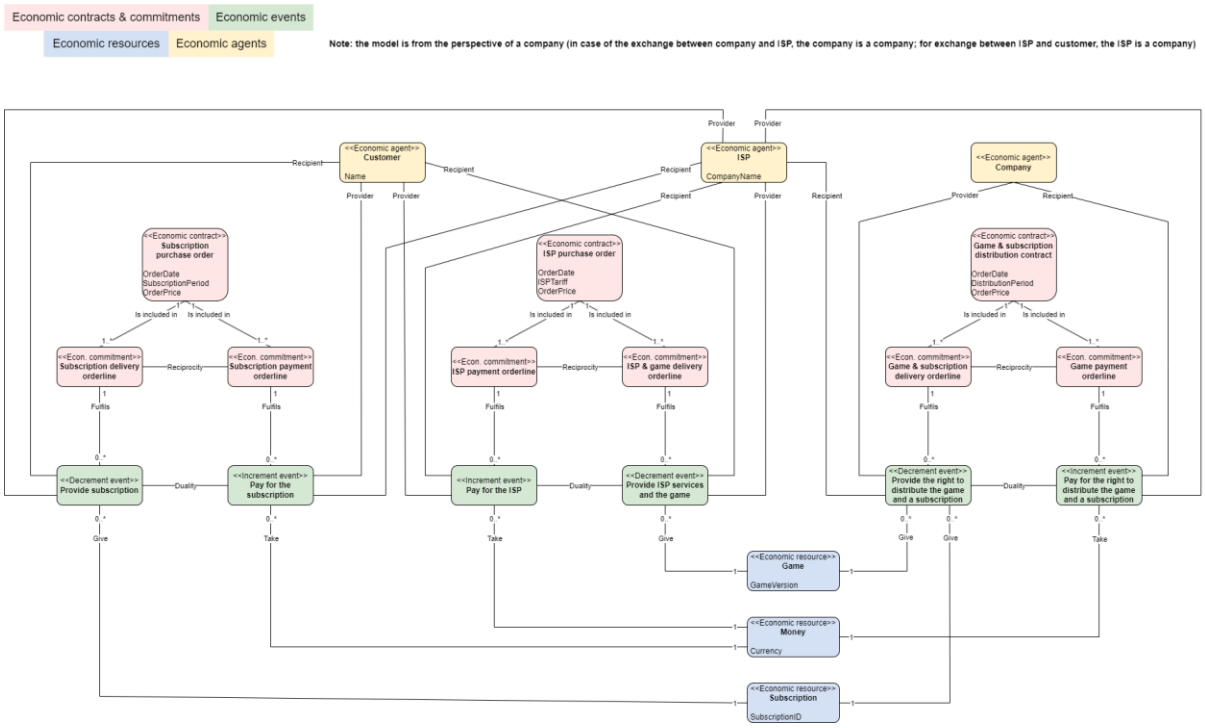
Influencers are also modeled to illustrate how external circumstances can be an opportunity or a threat as well as internal strengths and weaknesses.

4. Process Design

4.1 Exchange and Conversion Processes in REA

REA models provide a more detailed explanation of VDML exchange processes.

4.1.1 Between the Customer, ISP, and Company

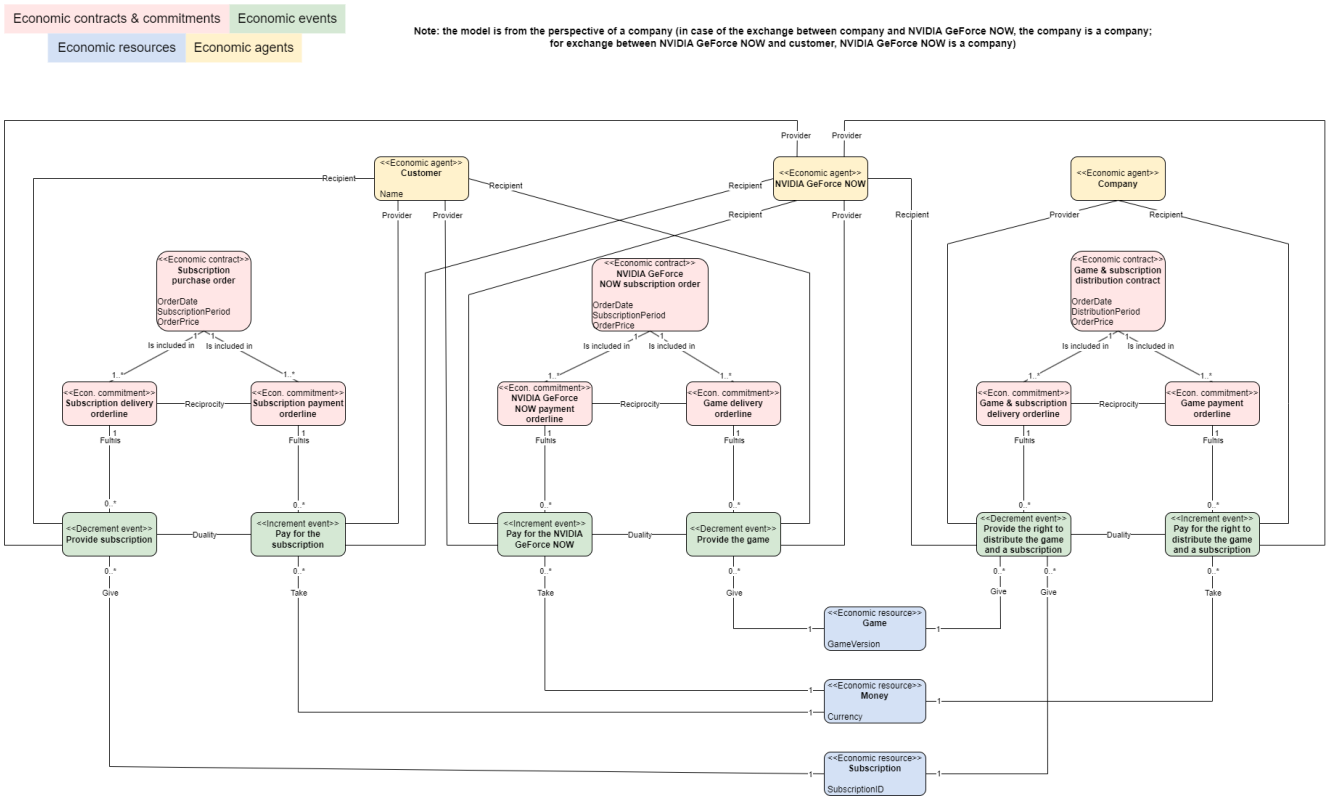


ISPs provide a distribution of digital copies of the game as well as subscriptions. The company does not sell the game directly to the customers, so they need to use ISP or Nvidia GeForce NOW to get it. For customer convenience, ISPs also provide subscriptions (customers do not need to go anywhere else, they can buy subscriptions directly from ISP). To illustrate this, REA diagram was divided in 3 parts:

1) Exchange of game and subscription distribution rights for money between the company and ISP
2) Exchange of ISP services and game for money between ISP and customers
3) Exchange of subscription and money between ISP and customers

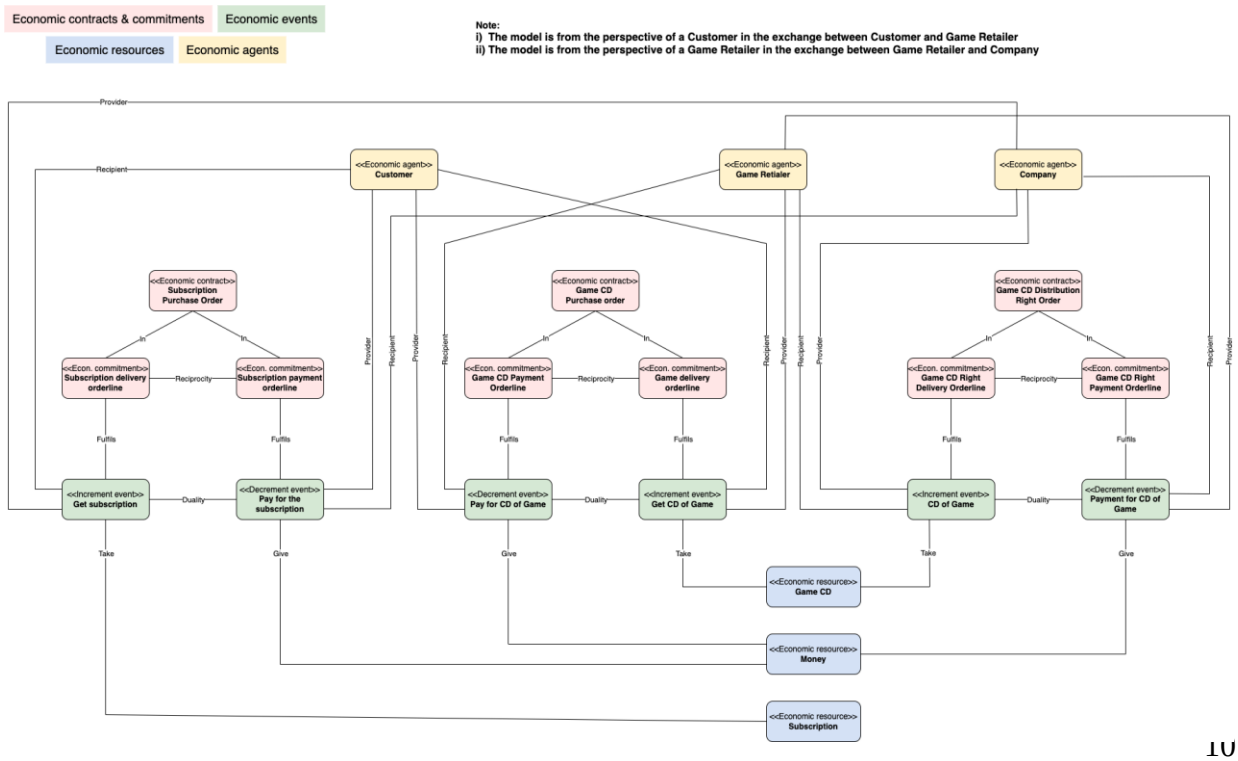
Each exchange has a dedicated contract (with date, price, and period (tariff for ISP) attributes) and commitments. Economic resources that are exchanged are: game (meaning a digital copy of the game, has attribute game version), money (can be in different currencies), subscription (with unique subscription ID to prevent piracy)

4.1.2 Between the Customer, NVIDIA GeForce NOW, and Company



The exchange processes involving Nvidia GeForce NOW are exactly the same as for the ISP. Same as ISP, Nvidia GeForce NOW distributes digital copies of the game and subscriptions.

4.1.3 Between the Customer, Game Retailer, and Company



This REA model has been designed from the perspective of the customer first, the game retailer then and the game company next. There are three economic exchanges between the customer, game retailer, and game company. Customers can buy gaming CDs from the game retailers and enjoy the game till the free trial ends, then they can take subscriptions from the company. On the other hand, game retailers pay the company and get the rights to distribute and sell the game CD to the customers. Details of these exchange processes are described below.

1. **Exchange between Customer and Game Retailer:**
    - **Economic Contract**: Subscription Purchase Order.
        - **Parties**:
            - Customer
            - Our Company
        - **Commitments**:
            - The Customer commits to paying for the subscription
            - The Game Retailer commits to delivering the subscription
        - **Decrement Event**: The Customer pays money for the subscription
        - **Increment Event**: The Customer receives the subscription service from the company

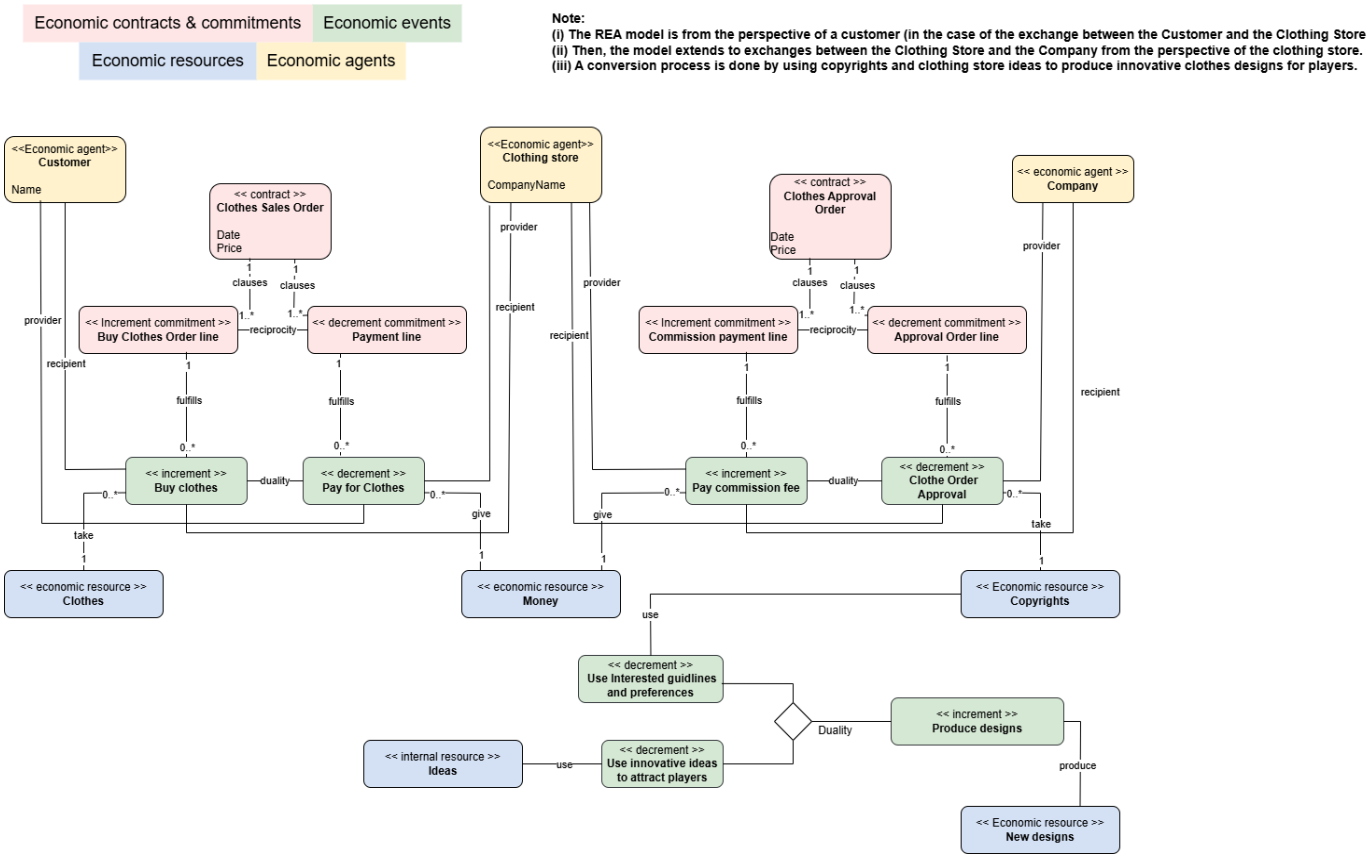
2. **Exchange between Customer and Game Retailer (Game CD):**
    - **Economic Contract**: Game CD Purchase Order.
        - **Parties**:
            - Customer
            - Game Retailer
        - **Commitments**:
            - The Customer commits to paying money to the Game Retailer for the Game CD
            - The Game Retailer commits to delivering the Game CD to the customer
        - **Decrement Event**: The Customer pays money to the Game Retailer for the Game CD
        - **Increment Event**: The Customer receives the Game CD from the Game Retailer

3. **Exchange between Game Retailer and Company (Game CD Distribution Rights):**
    - **Economic Contract**: Game CD Distribution Right Order.
        - **Parties**:
            - Game Retailer
            - Our Company
        - **Commitments**:
            - The Game Retailer commits to paying money to the Company for the right to distribute the Game CD

- The Company commits to delivering the distribution rights for the Game CD to the retailer

- **Decrement Event**: The Game Retailer pays money to the Company for the distribution rights
- **Increment Event**: The Game Retailer receives the right to distribute the Game CD from the company

4.1.4 Between the Customer, Clothing Store, and Company



The REA model is from the perspective of a customer in the case of the exchange between the Customer and the Clothing Store. Customers buy standout clothes from the Clothing store. To get branded clothing, the model extends to exchanges between the Clothing Store and the Company from the perspective of the clothing store. Clothing store provides a commission fee to our company for branded merchandising. There's a conversion process that uses copyrights and ideas provided by the company. It produces new designs that attract customers and motivate them to play actively by using the preferences, guidelines and innovative ideas.

a) Exchange between Customer and Clothing Store:

- **Economic Contract**: Clothes Sales Order.

- **Commitments**:
  - The Customer commits to buying branded clothes.
  - The Clothing Store commits to accepting payment for the clothes.
- **Decrement Event**: The Customer pays for the clothes.
- **Increment Event**: The Customer receives the clothes.

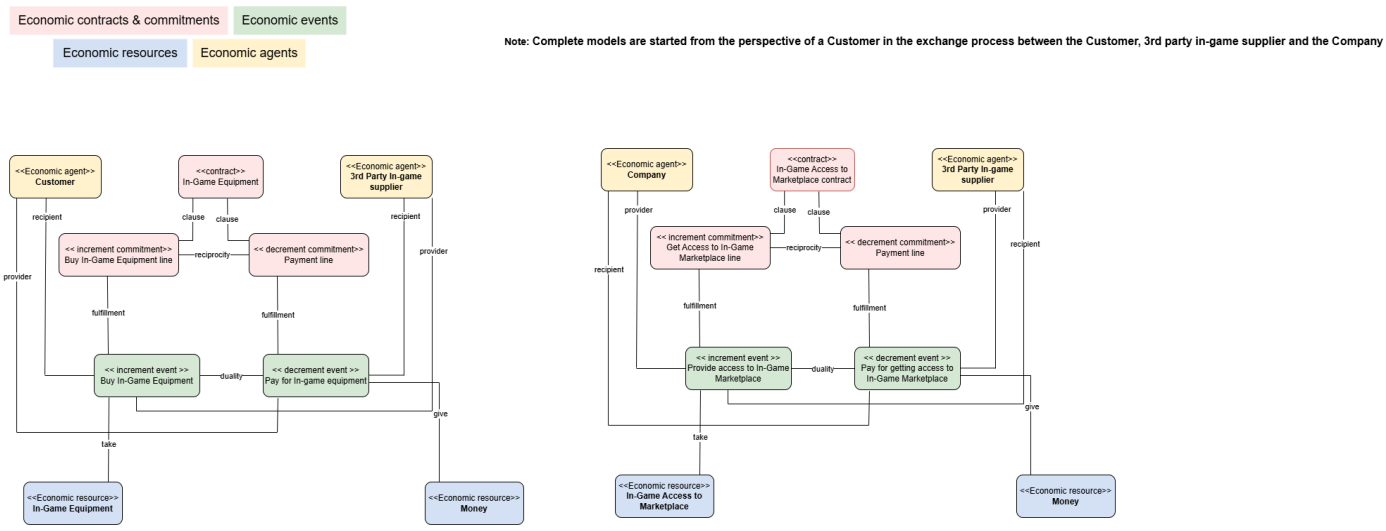
b) Exchange between Clothing Store and Company (Approval for Clothes Design):

- **Economic Contract**: Clothes Approval Order.
  - **Commitments**:
    - The Clothing Store commits to paying the Company a commission fee.
    - The Company commits to approving the clothing design order.
  - **Decrement Event**: The Clothing Store pays the commission fee to the Company.
  - **Increment Event**: The Company approves the clothing order.

c) Conversion Process within the Clothing Store

- The Clothing Store uses **Copyrights,** and **internal resources** like **Ideas** to create innovative designs.
  - **Decrement Events**:
    - Use interesting guidelines and preferences from copyrights.
    - Use innovative ideas to attract players
  - **Conversion Process**:
    - The Clothing Store uses these ideas to produce **designs for clothes that motivate players towards the game**.
  - **Increment Event**: The new designs are produced, adding to the Clothing Store's internal resources as **New Designs**.

4.1.5 Between the Customer, 3rd Party In-Game Supplier, and Company



a) Exchange between Customer and 3rd Party In-Game Supplier (In-Game Equipment):

In this REA (Resource-Event-Agent) model, we have two main players: the Customer and a Third-Party In-Game Supplier. They interact through a contract called In-Game Equipment, which outlines the exchange of virtual goods (in-game items) for money.

The model revolves around two key resources: In-Game Equipment and Money. To support this exchange, there are two commitments in place: the Buy In-Game Equipment Line (representing the customer's promise to buy) and the Payment Line (representing the customer's promise to pay). These commitments guide both parties in fulfilling their roles in the transaction.

Once the commitments are fulfilled, two events capture the actual exchange: Buy In-Game Equipment and Pay for In-Game Equipment. Here, the Customer is the recipient of the in-game equipment and the provider of the payment. Meanwhile, the Third-Party Supplier is the provider of the equipment and the recipient of the payment.

The perspective of this REA model is centered around the Customer, reflecting the inflow of in-game equipment as an increment and the outflow of payment as a decrement.

b) Exchange between Company and 3rd Party In-Game Supplier (In-Game Marketplace Display):

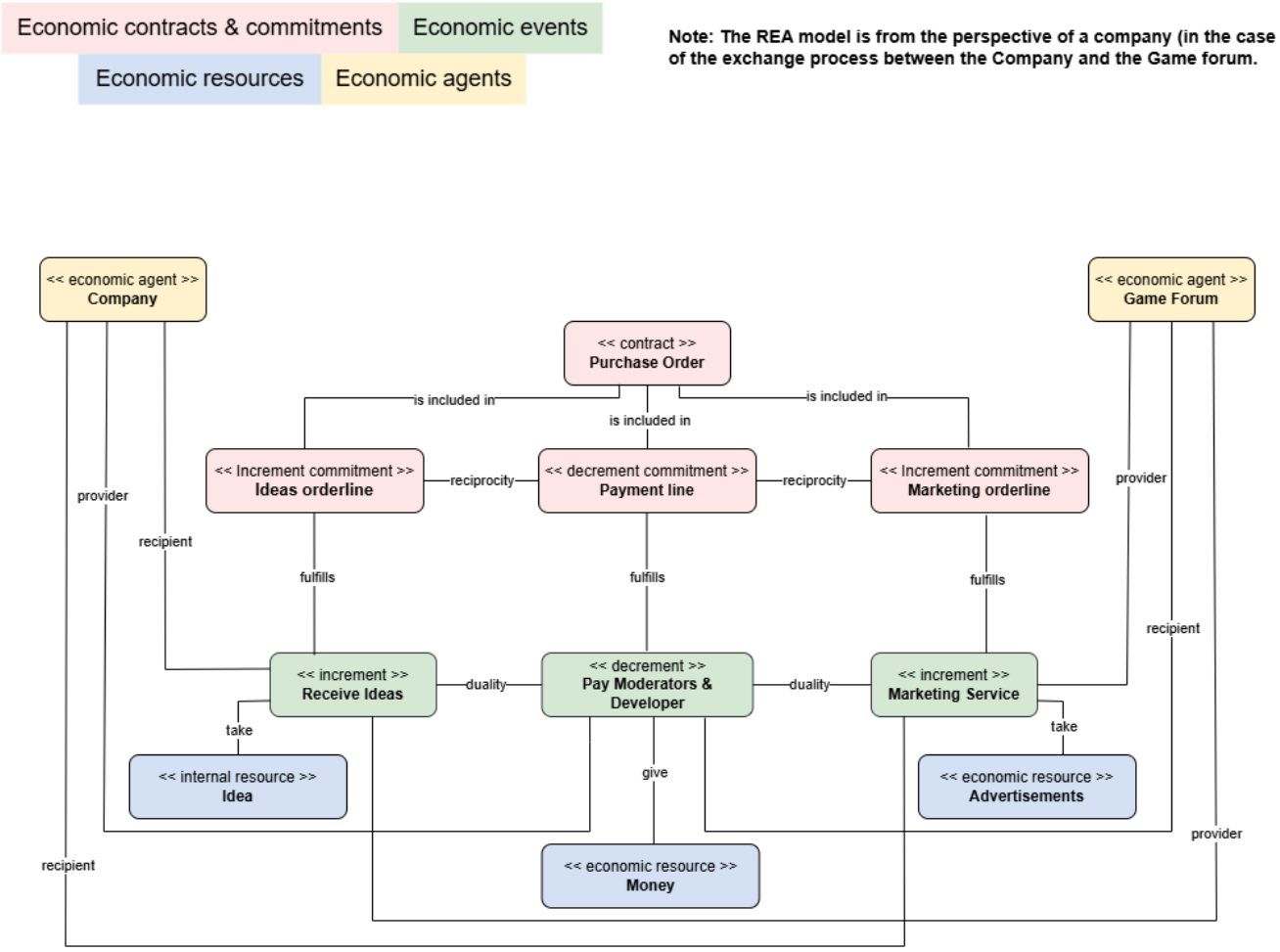
In this REA (Resource-Event-Agent) model, we examine the interaction between a Company and a Third-Party In-Game Supplier, governed by a contract: the In-Game Access to Marketplace Contract. These agreements define the exchange of two economic resources: Money and In-Game Access to the Marketplace.

The model includes two key commitments: Get Access to In-Game Marketplace Line (increment commitment) and Payment Line (decrement commitment). These commitments reflect the obligations between the Company and the Supplier, where the Supplier commits to paying for the marketplace access, while the Company commits to providing access to the marketplace.

Two corresponding events capture the execution of these commitments: Provide Access to In-Game Marketplace (increment event) and Pay for getting Access to In-Game Marketplace (decrement event). From the Supplier's perspective, they are the provider of Money and the recipient of Access to Marketplace. In contrast, the Company is the provider of Access to In-Game Marketplace and the recipient of Money.

The perspective of this REA model is centered around the supplier, reflecting the inflow of Access to In-Game Marketplace as an increment and the outflow of payment as a decrement.

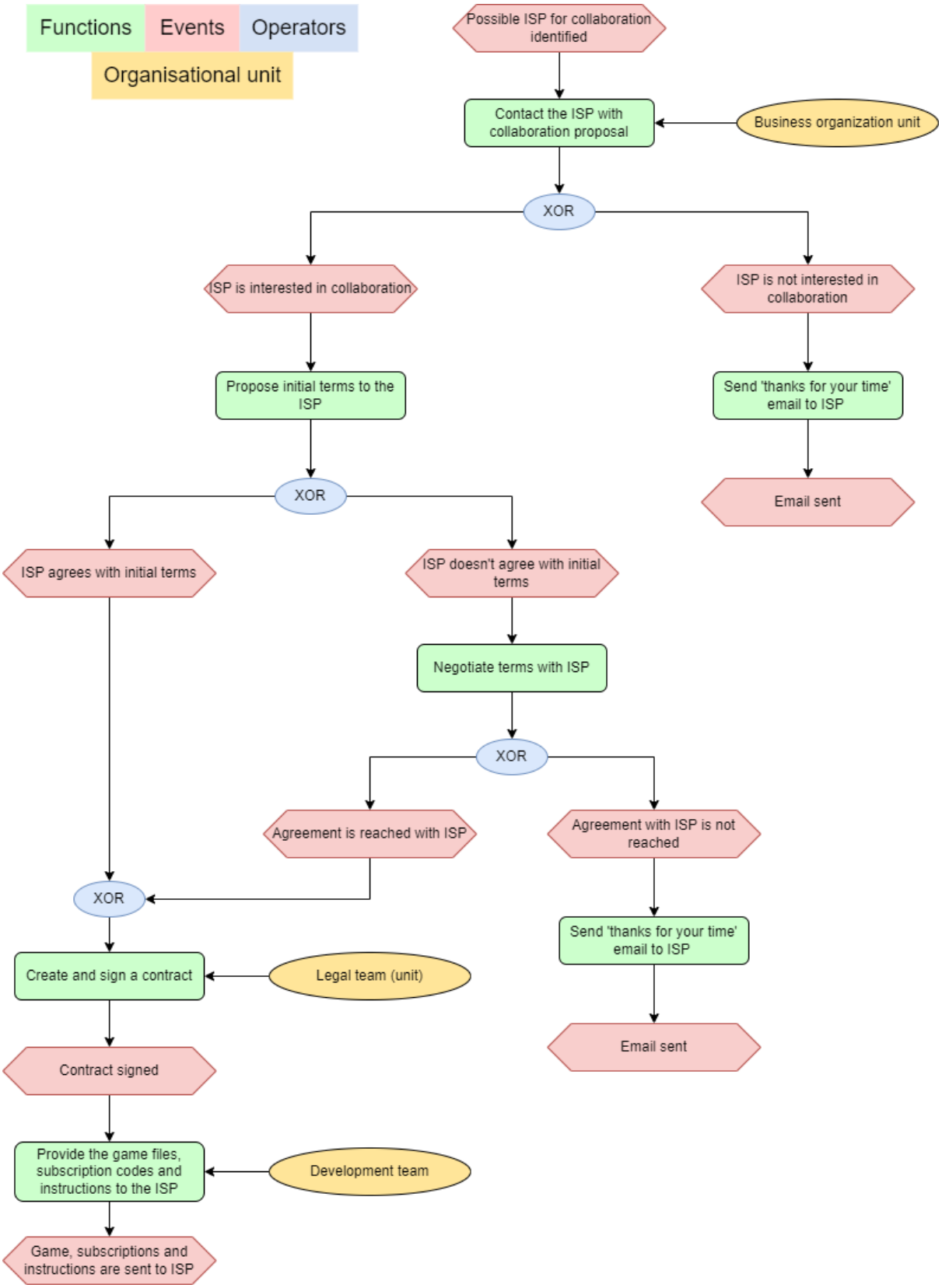
4.1.6 Exchange between Company and Game Forum:



- **Economic Contract**: Purchase Order.
  - **Commitments**:
    - The **Company** commits to paying the contributors of the **Game Forum** (e.g., paying moderators and developers).
    - The **Game Forum** commits to:
      - Sharing ideas to improve game performance.
      - Helping the Company with marketing the game.
  - **Decrement Event**: The Company pays the contributor of the Game Forum (e.g., for moderators and developers).
  - **Increment Events**:
    - The Company receives ideas from the Game Forum to improve game performance.
    - The Company receives marketing services like advertisements from the Game Forum.

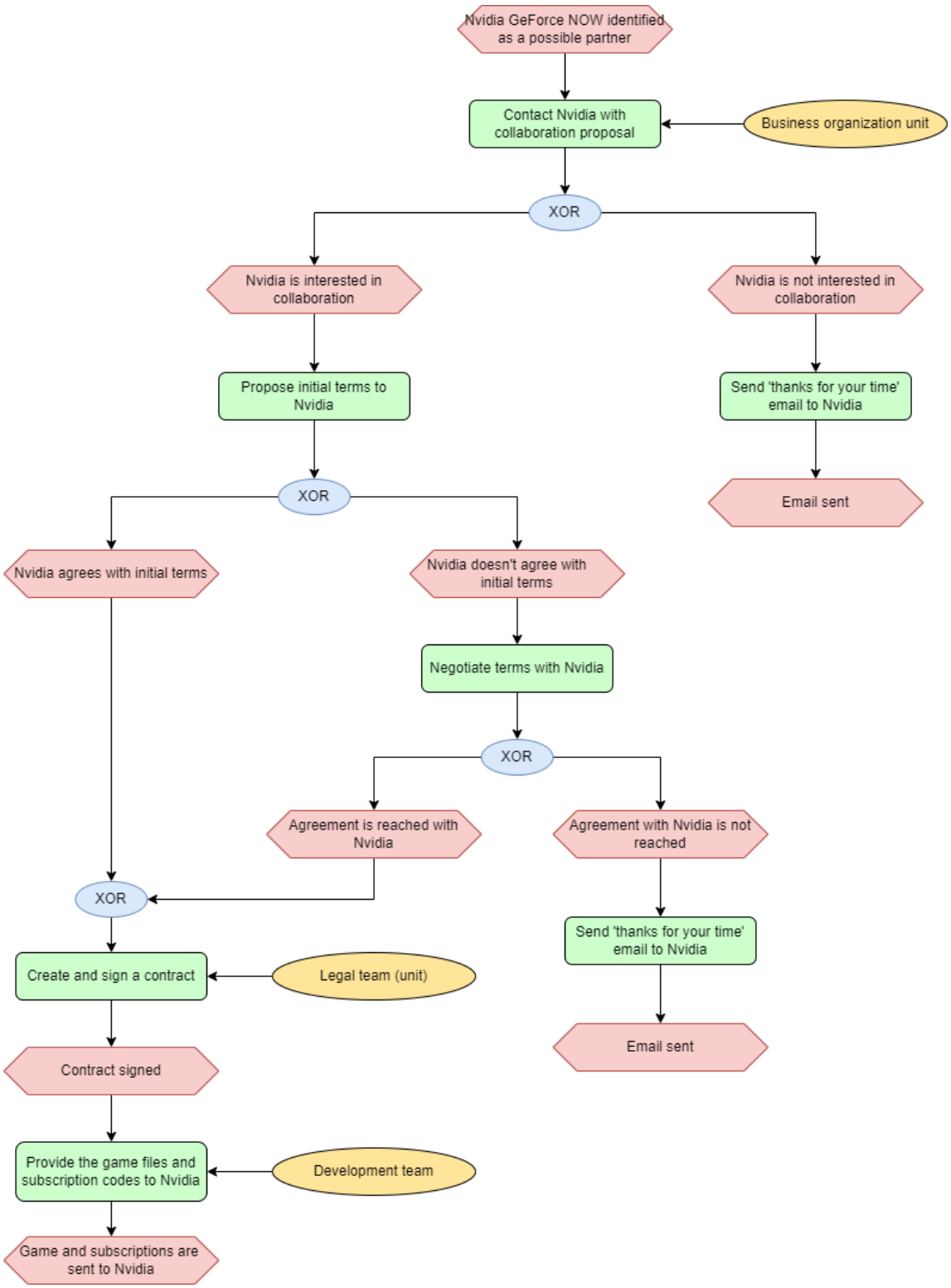
4.2 Detailed Processes in EPC

1. ISP Collaboration Process (Company & ISP)



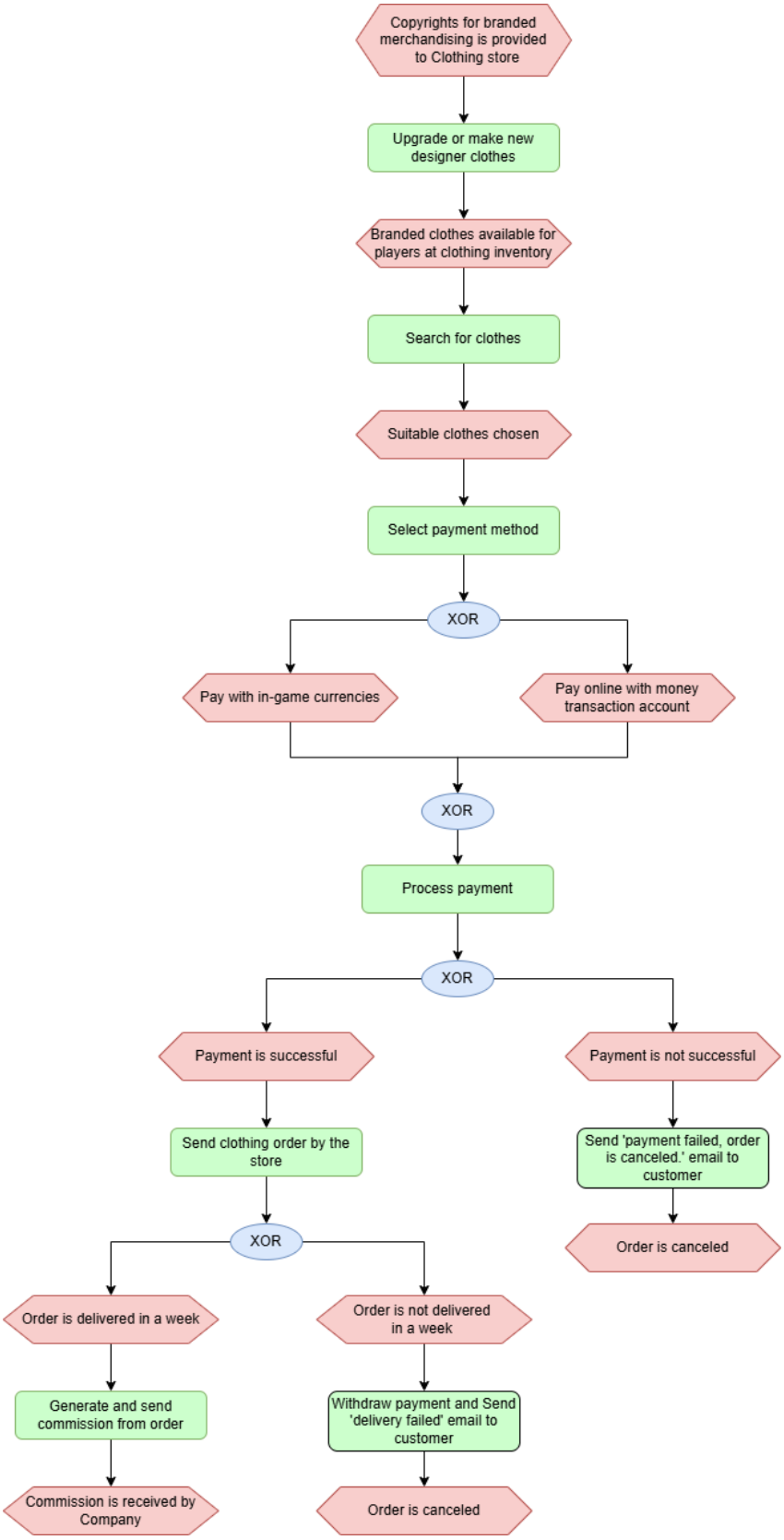
ISP Identification: The company identifies a potential ISP for collaboration.
Contact Proposal: A collaboration proposal is sent to the ISP by the business organisation unit.
Initial Terms Discussion: The ISP reviews initial terms and either shows interest or declines.
Negotiation: If ISP is not satisfied with terms but is interested in collaboration, terms are negotiated, and game files, subscription codes, and instructions are provided.
Contract Signing: If an agreement is reached, a contract is created by and signed by the legal team.
Providing the Game: After signing a contract development team provides ISP with the game, subscription codes and instructions.
Alternative Exit Processes: If no agreement is reached at any point, a "thanks for your time" email is sent and the process stops.

2. Nvidia GeForce NOW collaboration process (Company and Nvidia)



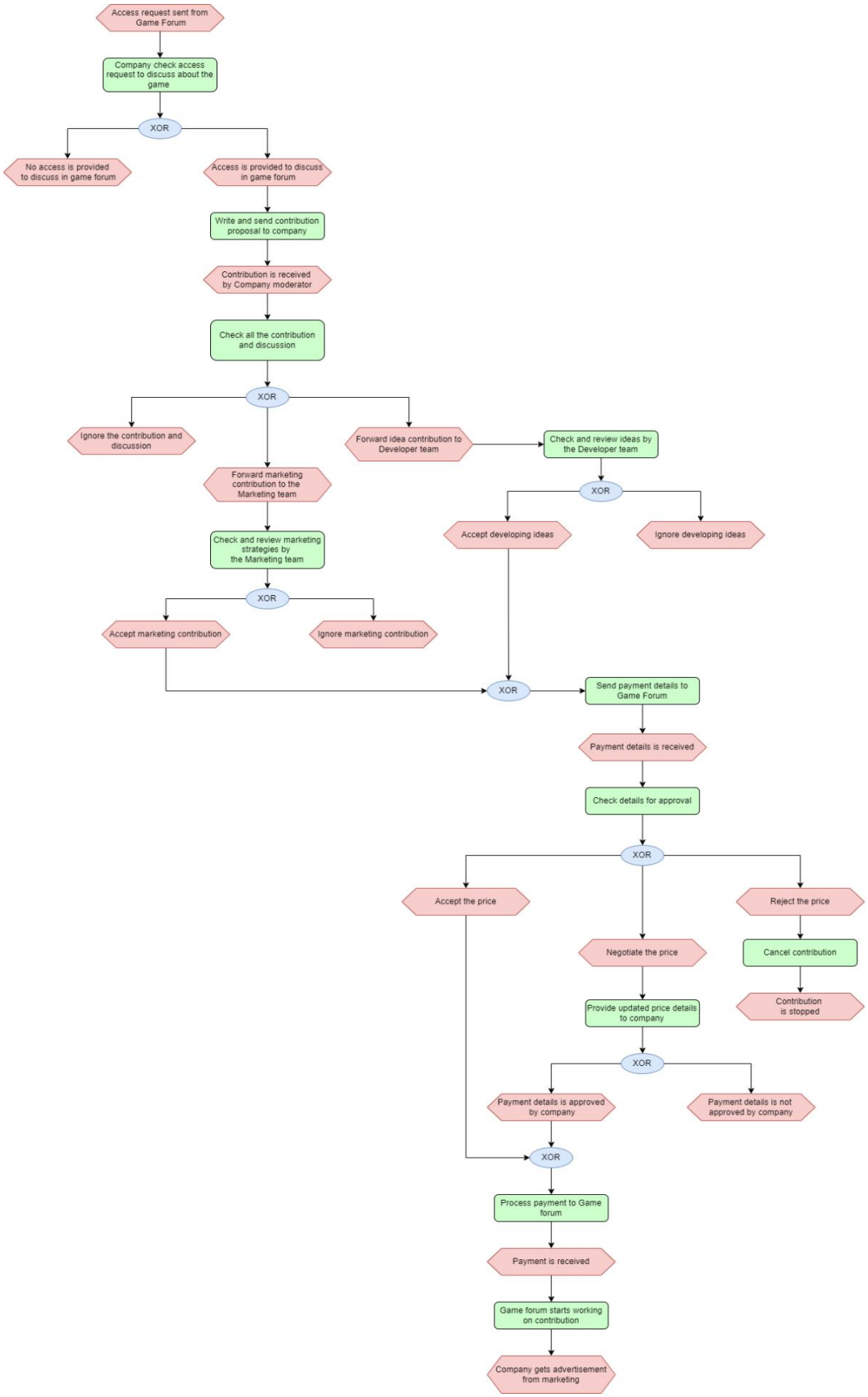
Collaboration with Nvidia to feature our game of Nvidia GeForce NOW follows the same steps as collaboration with ISP, only Nvidia does not need instructions from the development team as they have a lot of previous experience in integrating different games (they only require game and subscription codes).

3. Branded Clothing Order Process (Company -> Clothing Store -> Customer)



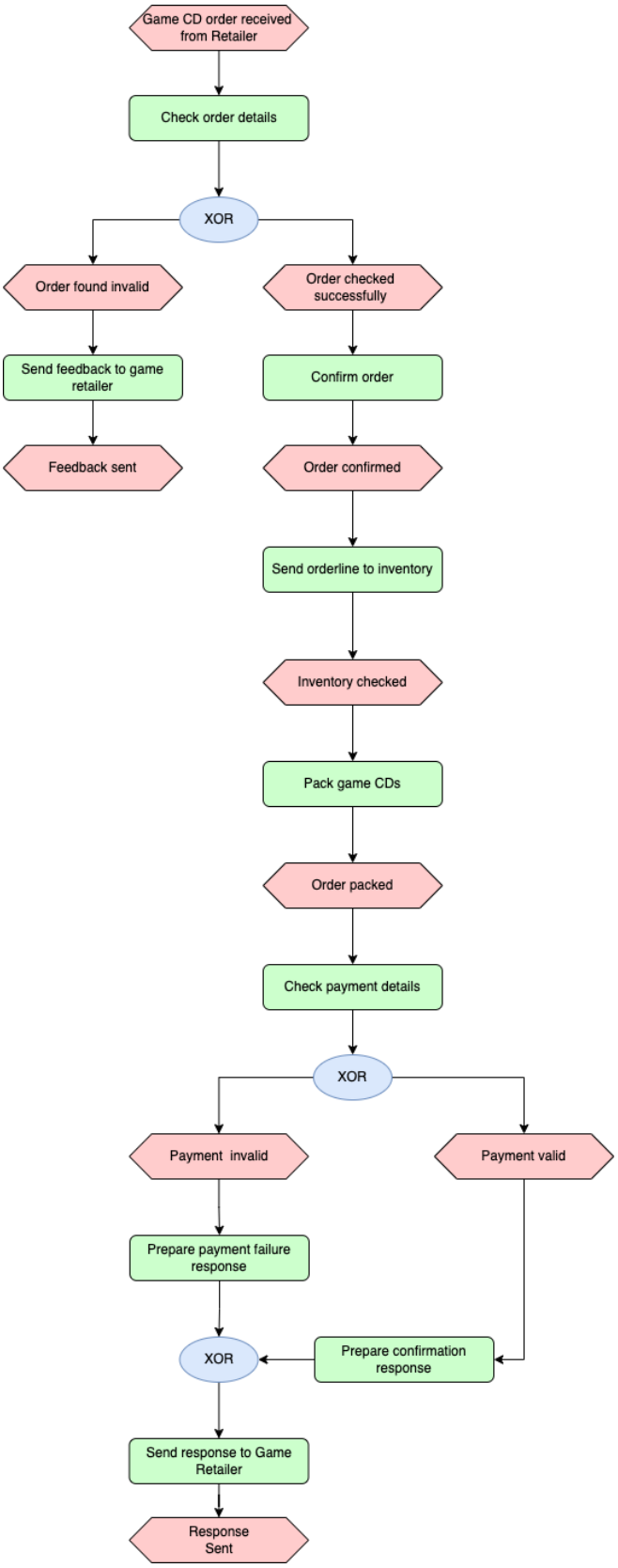
- **Product Creation using copyrights:** The clothing store receives copyrights for branded merchandising, which allows them to upgrade or create new designer clothing items.
- **Inventory Update:** Once designed, the branded clothes become available in the clothing store's inventory for players to browse.
- **Item Selection and Payment Choice:** Customers search the inventory for suitable clothes, select items they wish to purchase, and choose a payment method. They can either pay with in-game currency or use an online money transaction account.
- **Payment Processing:** The selected payment method is processed. If payment is unsuccessful, the customer is notified, and the order is canceled.
- **Order Fulfillment:** If payment is successful, the clothing store sends the order to the customer.
- **Delivery Status and Follow-up:** If the order is delivered within a week, a commission from the sale is generated and sent to the company. If the order is not delivered within a week, the payment is withdrawn to customer's account, a "delivery failed" email is sent to the customer, and the order is canceled.

4. Game Forum (Marketing & Development Ideas)



- **Get access to Contribute to the Game Forum**: The Company gives access to the game forum to the people who are part of the game.
- **Comments and contributions in the forum**: The company moderators go through and review every comment and contribution in the game forum.
- **Filtering and Forwarding:** Moderators forward the filtered ideas to the company and then the company decides which idea to accept and which one to ignore. After that, the company forwards the chosen contributed ideas to the marketing and development department separately.
- **Payment Processing**: The marketing and Development team gives their prices to the company and they can also negotiate if the company doesn't agree with the price.
- **Order Processing and Completion**: Once payment is successful, the company gets the idea to improve the game from the development team and gets advertisements from the marketing team.

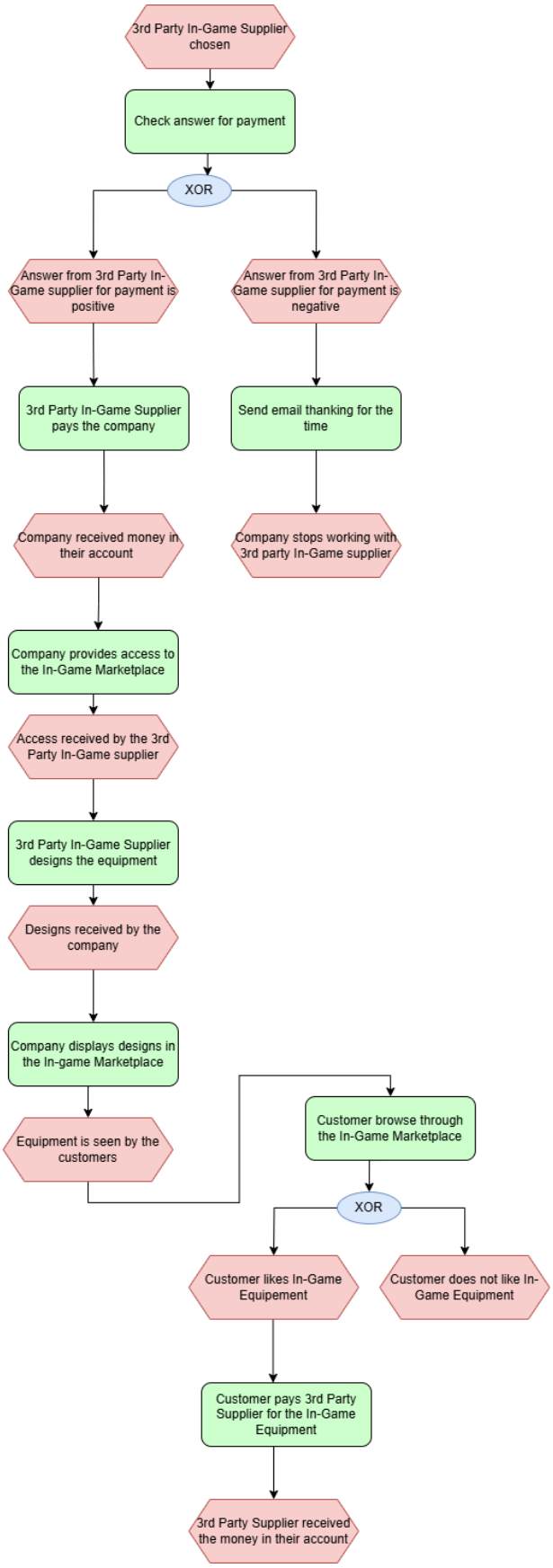
5. Game CD Order Process (Game Retailer)



In the economic process of ordering game CD by the game retailers following steps are involved:

- **Process Starting**: Process starts when a game CD order is received from the retailer
- **Order Details Checking**: The order details are checked. If the order information is valid, the process continues; if not then a feedback is sent to the retailer.
- **Order Confirmation**: The order is confirmed upon the successful checking of the details of the order
- **Inventory Coordination**: Once the order is confirmed an order-line is sent to inventory for stock confirmation.
- **Packing**: Once the Game CD is picked from the inventory it's packaged properly
- **Payment Verification**: Payment details are checked. If valid, a confirmation response is prepared; if not, a failure response is prepared.
- **Response to Retailer**: A response is sent to the retailer, either confirming or indicating a payment issue.

6. In-Game Equipment Purchase Process (Customer & 3rd Party Supplier)



- **Process Starting**: The process begins when a Third-Party In-Game Supplier is selected to participate in the marketplace.
- **Payment Verification**: The response regarding payment is checked. If the Third-Party Supplier responds negatively regarding payment, an email is sent to thank them for their time, and the Company discontinues the relationship. If the response is positive, the process continues.
- **Receipt of Payment**: Upon a positive response, the Third-Party Supplier pays the Company, and the Company verifies the receipt of funds in its account.
- **Marketplace Access Provision**: Once payment is confirmed, the Company provides access to the in-game marketplace for the Third-Party Supplier, and the access is acknowledged as received.
- **Design Creation**: The Third-Party Supplier begins designing the in-game equipment, which is then submitted to the Company.
- **Display in Marketplace**: The Company displays the newly designed equipment in the in-game marketplace, making it visible to customers.
- **Customer Interaction**: Customers browse the in-game marketplace, viewing the equipment.
- **Customer Decision**: If customers do not like the equipment, the browsing session ends. If they like the equipment, the process continues with the customer purchasing the in-game equipment.
- **Final Payment**: The customer pays the Third-Party Supplier for the equipment, and the Third-Party Supplier confirms receipt of the payment in their account.

4.3 Relationships between Models

Goals	REA	EPC	VDML
Establishing collaboration with third-party suppliers.	The company gives third parties access to the in-game marketplace and makes a profit.	The third parties sent the proposal to the company if the company agreed with terms they give access to use in-game marketplace	Third parties pay value proposition "money" to the company and get value proposition access to the "in-game marketplace".
Game experience improvement.	The company gives money to the developer to improve the game performance and get ideas from the game forum to work with those ideas to improve the game experience.	The company sends filter ideas from the game forum to the developer with payment proposals, if the developers agree with the payment they work with those ideas to improve the gaming experience.	The company pays the game forum's contributors value proposition "money" and gets value proposition "Idea" to improve gaming experience for the user.
Increase revenue from brand merchandising	The company gives "clothes sales order" approval to the clothing store and by selling clothes, it fulfills the "pay a commission fee" order line to get revenue.	It illustrates the processes to earn revenue such as "Branded clothes available for players at clothing inventory" to sell the clothes and get a commission.	It shows how "copyright" gives the right to sell standout clothes and "money" as a value increases revenue in the company.

1. Customer and 3rd Party Supplier Relationships

- **Buying In-Game Equipment** and **Ordering Game CDs**:
  - In both processes, the **Customer** buys something from a third-party seller: either virtual items (like in-game equipment) or physical items (like game CDs).
  - The steps are similar: the customer chooses a product, pays, and then receives the item.

- Both processes show how customers interact with outside sellers to get the products they want.

2. Company and 3rd Party Supplier Relationships

- **In-Game Equipment** and **ISP Collaboration**:
  - In these processes, the **Company** works with outside partners to offer more products or services.
  - For in-game equipment, the Company lets 3rd party suppliers sell their items in its marketplace, adding more variety for players.
  - For ISP collaboration, the Company works with internet providers to make its games and subscriptions available to more people.
  - These partnerships help the Company reach more customers and offer more options without having to create everything themselves.

3. Merchandise (Branded Clothing) and Customer Purchase Behavior

- **Branded Clothing** and **In-Game Equipment**:
  - Both involve customers buying extra items, either physical (like branded clothes) or virtual (like in-game gear).
  - The processes are similar: customers choose what they like, pay, and then receive the item.
  - Both types of items (clothing and in-game equipment) are ways for the Company to keep players engaged and offer things they enjoy.

4. Game Retail and ISP Collaboration Relationship

- **Ordering Game CDs** and **ISP Collaboration**:
  - Both involve distributing the Company's games, but through different methods.
  - For Game CDs, the Company sells physical copies through retailers.
  - For the ISP collaboration, the Company provides digital access to games through an internet provider.
  - Together, these two methods help the Company reach different types of customers: those who want a physical product and those who prefer digital access.

5. Internal Relationships within the Company

- **In-Game Equipment** and **Branded Clothing**:
  - Both show how the Company provides a platform (or space) for outside products.
  - The Company lets third parties offer virtual items (in-game equipment) or physical products (branded clothing) to its customers.
  - This adds more variety for customers, helps the Company make extra income, and strengthens its brand by offering related products.

4.4 Operational Modules

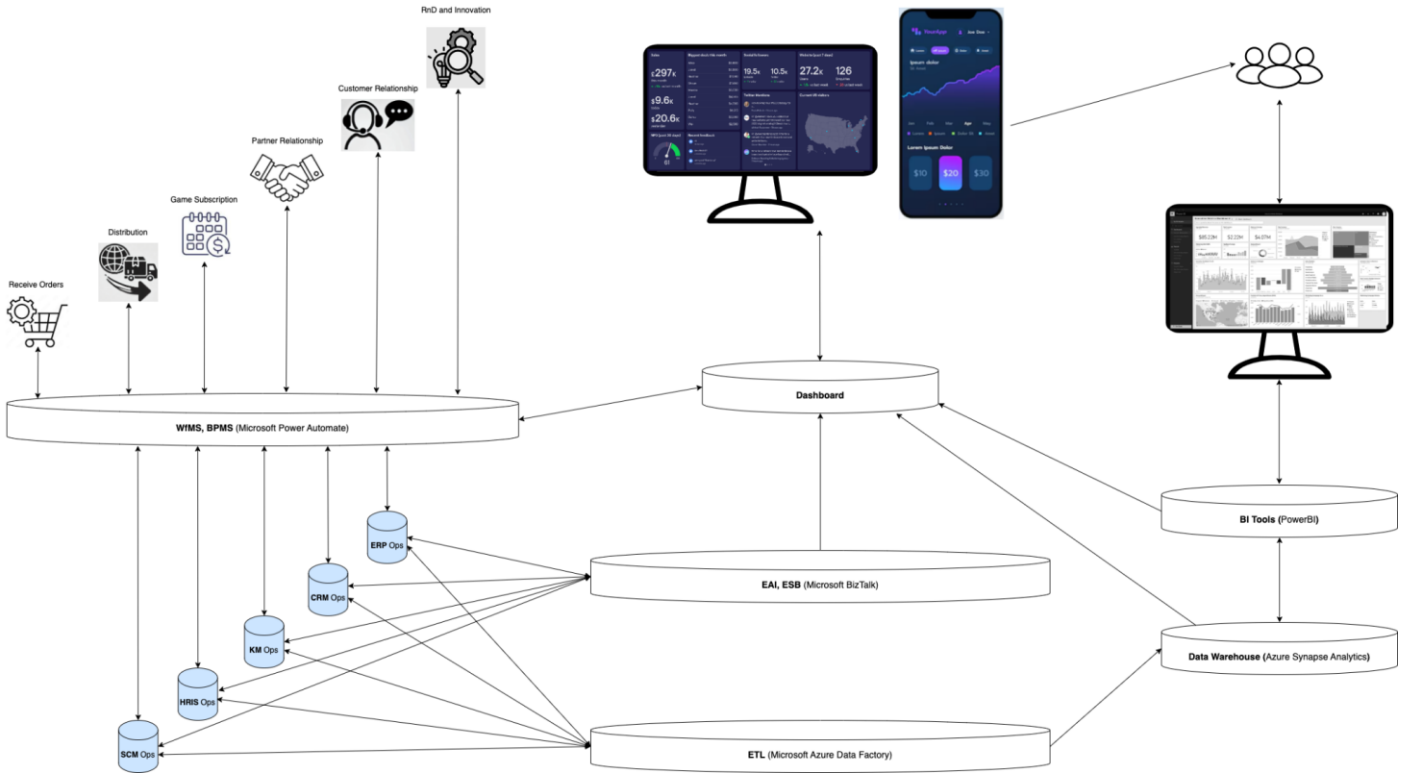
Our gaming company includes different operational modules to ensure the best gaming experience of its kind. We execute this by breaking the operations of the company in different operational departments or modules such as:

- Order collection and Sales
- Game distribution
- Subscription management
- Partner relationship management
- Customer relationship management
- Innovation, Research and Development

We internally use different tools to optimize the operation and performance of the department. Each department can independently choose their preferred tools or platforms as long as they can be integrated to our IT Architecture through necessary interfaces. For example, the sales team and the customer service team prefer to use Salesforce whereas the Innovation and RnD team prefers to use GitHub, Slack, and other tools.

All these tools are connected through Microsoft BizTalk, our Enterprise Application Interface. They exchange data and communicate with each other.

5. IT Architecture Design



5.1 IT Architecture Components

5.1.1 WfMS & BPMS (Microsoft Power Automate)

Workflow Management System manages and automates workflows for various business processes such as order and sales management, distribution management, subscription management, and customer service. We've used Microsoft Power Automate that helps streamline tasks across departments, reducing manual intervention and improving process efficiency.

Microsoft Power Automate was chosen over alternatives like IFTTT, Zapier, Nintex, Appian, or Zoho Creator, due to its seamless integration with other Microsoft products and its reliability, and enterprise-grade automation capabilities, which align well with the project's need for efficiency and scalability in managing workflows.

5.1.2 EAI & ESB (Microsoft BizTalk)

We've used Enterprise Application Integration (EAI) which routes, transforms, and orchestrates communication between the operations modules. It enables smooth data flow and integration between different applications and systems. In our company, we use Microsoft BizTalk as EAI to enable all the applications and tools to communicate with each other.

Microsoft BizTalk was selected over the other alternatives like Apache Camel, Apache Kafka, MuleSoft etc. due to its robust integration features, especially within Microsoft-centered IT environments. BizTalk's capabilities in managing data transformation and ensuring reliable integration align with the enterprise needs of the project.

5.1.3 ETL (Microsoft Azure Data Factory)

Our systems continuously generate a huge amount of data every moment. We capture these data and try to figure out the insights out of it. To do so we use the Extract, Transform and Learn (ETL) process. We use Microsoft Azure Data Factory as the ETL service. It collects all the structured and unstructured data from different sources, then processes it through the necessary data pipeline and store it in the data warehouse.

5.1.4 Data Warehouse (Azure Synapse Analytics)

Data Warehouse is used to store, process, and analyze large volumes of data coming from the ETL pipeline. Azure Synapse provides a centralized data repository for the MMORPG, enabling advanced analytics and reporting.

There are a number of options in terms of data warehouse, e.g. Amazon Redshift, Google BigQuery, Snowflake etc. We've chosen Azure Synapse because of its high performance, cost effectiveness and compatibility with other tools and services.

5.1.5 BI Tools (Power BI)

Business Intelligence tools are used to query, analyze and visualize data, creating interactive dashboards and reports for real-time insights into business operations. It enables stakeholders to monitor KPIs and make data-driven decisions. In our company we use Microsoft Power BI as the business intelligence platform.

We've chosen Power BI over other alternatives in the market, for example: Tableau, Qlik Sense, or Looker, because of its deep integration with other Microsoft services, intuitive interface, and powerful analytics capabilities. It provides easy access to visual data insights for stakeholders, ensuring that metrics and KPIs are visible across the organization.

5.1.6 Dashboards (Custom)

We create custom dashboards to present the key insights of the overall performance of the company. The management and the key board members can view necessary insights directly from the dashboard.

5.2 Data Flow

In our IT architecture the services and tools used in the operational modules work independently which makes the operational teams to focus only on their own activities instead of learning new tools and technologies. Then we put in a Workflow Management System (WfMS) layer which enables us to add any new activities/actions to the operational flow without changing their own workflow or without having the need of customization of the tools that department is using.

Let's give one example of how the WfMS works in our IT architecture: suppose we need to send an email when a ticket status changes in the Customer Relationship Management (CRM) software. We use Salesforce and we don't need to ask Salesforce to add this feature if we have the WfMS in place. We can just open a new workflow which subscribes to the ticket status and takes necessary actions when the workflow is triggered.

The operational modules can exchange data to each other through the Enterprise Application Interface (EAI). And all the data these operational components generate are passed through our ETL pipeline and eventually stored to our Data warehouse. Finally, the data is fed to the Business Intelligence tool and visualized to the key stakeholders. Besides, we've a custom dashboard that supports mobile friendly views to get quick insights for the key stakeholders.

6. Business Performance Management

6.1 Leading and Lagging KPIs

Lagging KPIs:

Name	Definition	Related to goal	Target value	Measures business process	IT system data source	IT solution feeding the dashboard
Total Profit	Total monthly revenue	Increase Profitability	$500,000 per month	Revenue generation	ERP Ops	Bi Tools
Daily Active Users	Measures the number of unique players logging into the game daily.	Increase Player Retention and Engagement	10% monthly growth	Engagement levels	CRM Ops	Data Warehouse
Security Incident Rate	Monthly rate of security incidents affecting gameplay.	Ensure Information Security and Compliance	Fewer than 2 incidents per month	Security compliance	WIMS	BI tools

Leading KPIs:

Name	Definition	Related to goal	Target value	Measures business process	IT system data source	IT solution feeding the dashboard
Monthly Average Player Level Progression	Average level progression achieved by players monthly.	Increase Player Engagement	Level 5 average for active users per month	Gameplay activity	CRM Ops	BI Tools
Marketplace Transaction Volume	Total volume of items bought and sold on the in-game marketplace.	Boost Player Retention and Engagement	10% monthly increase	In-game trading activity	SCM Ops	EAI
Subscriptions Transaction Volume	Total volume of subscriptions bought.	Boost Player Retention and Engagement	100000 subscriptions per month	Subscription purchase	ERP Ops	Data Warehouse
In-Game Activity Completion Rate	Percentage of players completing daily quests and events.	Enhance Player Engagement and Experience	70% daily completion rate	Game activity engagement	CRM Ops	BI Tools

Customer Support Ticket Volume	Measures the number of support tickets related to bugs or feedback.	Improve Communication Between Players and Developers	Fewer than 50 tickets per month	User satisfaction and engagement	CRM Ops	Data Warehouse
Churn Rate	Percentage of players unsubscribing each month.	Increase Player Retention and Engagement	Below 5%	Retention	CRM Ops	BI Tools

6.2 Evaluation of KPIs

KPI	Aligned with goals	Easy to understand	Balanced	Standardised	Actionable	Owned	Accurate	Timely	Few in number
Total Profit	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Daily Active Users	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	
Security Incident Rate	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	
Monthly Average Player Level Progression	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Marketplace Transaction Volume	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Subscriptions Transaction Volume	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
In-Game Activity Completion Rate	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Customer Support	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	

Ticket Volume									
Churn Rate	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	

7. Enterprise Architecture Evaluation

7.1 Enterprise Architecture Modelling

The Enterprise Architecture (EA) model illustrates an organized flow across three main layers: Business Process, Application Process, and Technology Process.

1. Business Process Layer:

This layer focuses on the customer-company interaction and key business functions. It elaborates Customer Interactions that includes Order Intake, Order Processing, and Payment Processing as core operational functions. It also mentioned other strategic activities and goals that have an impact on IT architecture directly such as Data Security Compliance, Employee Management, and Operational Efficiency.

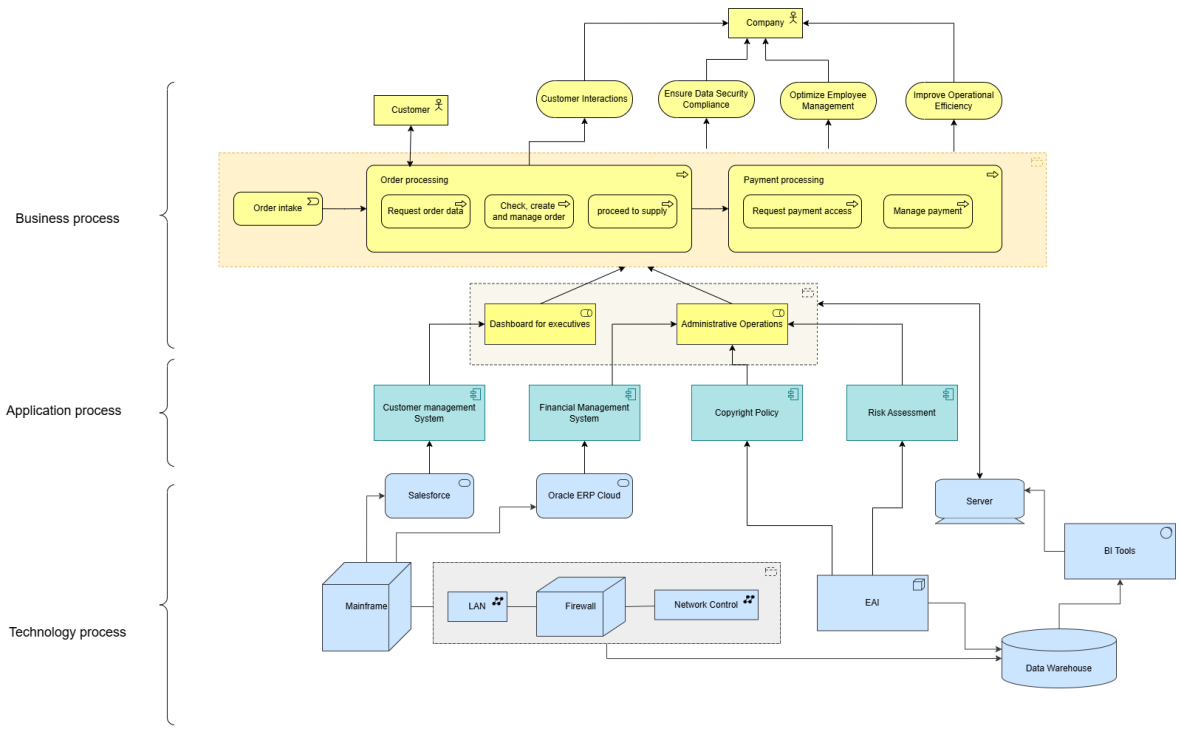
2. Application Process Layer:

This layer supports business processes through various applications, such as “Customer Management System” for handling customer-related data by integrating with Salesforce. Another application is “Financial Management System” that is linked with Oracle ERP Cloud for financial operations. Copyright Policy and Risk Assessment for regulatory, maintain confidentiality and risk management needs. Further, it provides Dashboards for Executives and Administrative Operations to enhance decision-making and administrative efficiency.

3. Technology Process Layer:

The application layer is then connected with essential IT infrastructure, including core systems like Mainframe, LAN, Firewall, Network Control, and EAI (Enterprise Application Integration) to facilitate secure and integrated data flows. Data Warehouse and BI Tools are used for data storage, analysis, and reporting, supporting decision-making and operational insights.

Finally, this layered approach emphasizes the integration between business processes, applications, and technology infrastructure to streamline operations and support strategic objectives within the organization.



7.2 Security Evaluation

For the drastic exposure of infrastructure to the internet and changing political situation, it becomes important to keep organizations secure. Some of the goals are threatened by serious security issues. Therefore, we need to analyze Security Vulnerabilities by using resources like the National Vulnerability Database (NVD) in our EA model ~~[1]~~[1]

- **Vulnerability Type: Server and Connection Security**
  - **Threat Description**: Insufficiently secured servers or unprotected data connections can cause data discrepancies, system outages, and interruptions to game operations.
  - **Impacted Goals**: Ensure Data Security Compliance; Improve Operational Efficiency.
- **Vulnerability Type: Protection of Customer Data and Privacy**
  - **Threat Description**: Lacking encryption can lead to data breaches, risking exposure of customer information and potentially damaging trust.
  - **Impacted Goals**: Ensure Data Security Compliance; Enhance Customer Trust and Interactions.
- **Vulnerability Type: Workflow and Process Security**
  - **Threat Description**: Unauthorized access to business workflows could delay processes, including subscription renewals or vendor approvals, leading to inefficiencies.
  - **Impacted Goals**: Improve Operational Efficiency; Optimize Employee Management.
- **Vulnerability Type: Financial Data Security**

    - **Threat Description**: Outdated ERP systems may leave financial information open to fraudulent activities, compromising financial data integrity.
    - **Impacted Goals**: Ensure Data Security Compliance; Improve Operational Efficiency.
- **Vulnerability Type: Access Management for Users (Dashboards/BI Tools)**
    - **Threat Description**: Poorly managed access permissions may enable unauthorized data access or alterations, which can lead to incorrect decision-making.
    - **Impacted Goals**: Optimize Employee Management; Enhance Customer Interactions through Data-Driven Insights.

[1] Ebbers, F., Hacks, S., Thakurta, R. (2021). The business impact of IIOT vulnerabilities. In PACIS 2021 Proceedings.